

Laplace approximation

Find the peak of the log posterior, measure how sharply it curves, and use the Gaussian with that centre and width.

THE QUESTION

The posterior $p(\theta | D) = p(D, \theta) / Z$ is known only up to the normaliser Z , but its peak is easy to find. Can a Gaussian at the peak stand in for it?

THE INTUITION

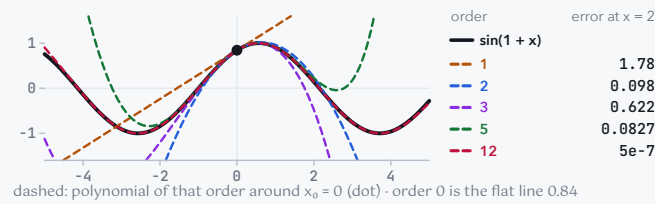
At the MAP the gradient of the negative log joint is zero, so nearby it looks like a parabola. The exponential of a parabola is a Gaussian, and a sharper peak gives a narrower one.

KEY TAKEAWAY

$p(\theta | D) \approx \mathcal{N}(\theta_{\text{MAP}}, H^{-1})$, with H the Hessian of $-\log p(D, \theta)$ at the MAP. It works for one peaked, nearly symmetric posterior and fails for skewed, bounded or two-peaked ones.

$p(D, \theta) \rightarrow f = -\log p(D, \theta) \rightarrow \text{minimise: } \theta_{\text{MAP}} \rightarrow H = \nabla^2 f(\theta_{\text{MAP}}) \rightarrow \mathcal{N}(\theta_{\text{MAP}}, H^{-1}) \rightarrow \text{sample, predict} \rightarrow Z \approx \text{evidence}$

1 A Taylor polynomial copies a function near one point



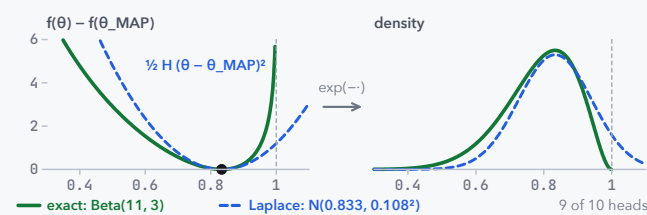
$$\tilde{f}(x) = \sum_{k=0}^K \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k$$

For $\sin(1+x)$ at $x_0 = 0$: $0.8415 + 0.5403x - 0.4207x^2 - 0.0901x^3 + \dots$ A low order is enough near x_0 ; at $x = 4$ order 5 is off by 5.05. In d dimensions, to second order, with $\delta = x - x_0$,

$$\tilde{f}(x) = f(x_0) + \nabla f(x_0)^T \delta + \frac{1}{2} \delta^T \nabla^2 f(x_0) \delta$$

The Hessian $\nabla^2 f$ holds $\partial^2 f / \partial x_i \partial x_j$ (autodiff or finite differences). For $\sin(1+x_1+x_2)$ at 0 every entry is $-\sin 1 = -0.841$: rank 1, so a Hessian can be singular.

2 Near the MAP, $-\log$ posterior is a parabola



With f the MAP training loss,

$$p(\theta | D) = e^{-f(\theta)} / Z, \quad f(\theta) = -\log p(D, \theta)$$

At θ_{MAP} the gradient is 0, so with $H = \nabla^2 f(\theta_{\text{MAP}})$

$$f(\theta) \approx f(\theta_{\text{MAP}}) + \frac{1}{2} (\theta - \theta_{\text{MAP}})^T H (\theta - \theta_{\text{MAP}})$$

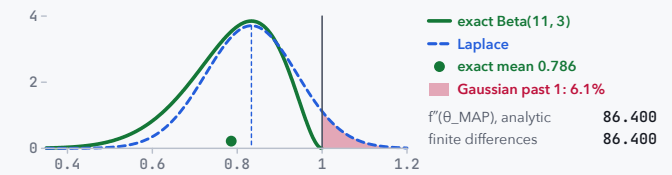
and the exponential of this parabola is a Gaussian:

$$p(\theta | D) \approx \mathcal{N}(\theta | \theta_{\text{MAP}}, H^{-1})$$

$$Z \approx p(D, \theta_{\text{MAP}}) (2\pi)^{d/2} |H|^{-1/2}$$

In 1D the variance is $1/f''(\theta_{\text{MAP}}) = -1/(d^2 \log p/d\theta^2)$. H must be positive definite. Inverting it costs d^3 for d weights, so large networks use diagonal or block approximations (the Laplace-Redux library).

3 Worked example: a coin with a Beta(2, 2) prior



data	posterior	MAP	Laplace sd	exact sd	past 1
9H 1T (notebook)	Beta(11, 3)	0.8333	0.1076	0.1059	6.1%
8H 1T (slide list)	Beta(10, 3)	0.8182	0.1163	0.1126	5.9%

Nine heads and one tail (the notebook's data) give the exact posterior Beta(11, 3). Up to a constant $f(\theta) = -10 \log \theta - 2 \log(1 - \theta)$, $\theta_{\text{MAP}} = 10/12 = 0.8333$

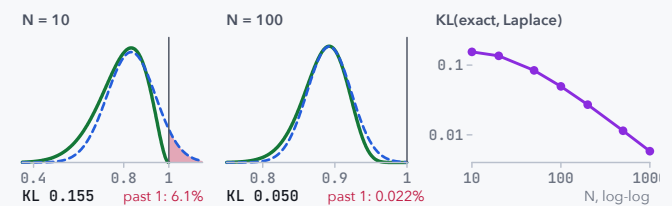
$$f''(\theta_{\text{MAP}}) = \frac{10}{\theta^2} + \frac{2}{(1-\theta)^2} = 14.4 + 72 = 86.4$$

so $\sigma = 1/\sqrt{86.4} = 0.108$. Central differences agree:

$$\frac{f(\theta + h) - 2f(\theta) + f(\theta - h)}{h^2} = 86.400, \quad h = 10^{-4}$$

The width is close to the exact 0.106, but the Gaussian is centred on the mode 0.833, not the mean 0.786, and 6.1% of it lies above 1.

4 More data makes the posterior more Gaussian

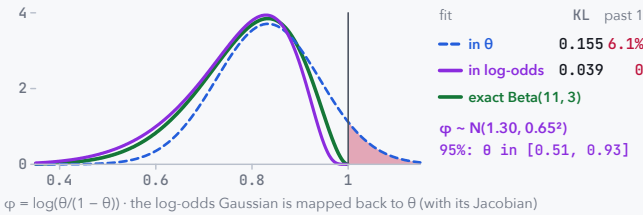


The log posterior is a sum of N log-likelihood terms, so the curvature grows like N and the sd shrinks like $1/\sqrt{N}$; the higher Taylor terms fall behind the quadratic one. At 90% heads the KL divergence from the exact posterior to Laplace goes from 0.155 at $N = 10$ to 0.050 at 100 and 0.006 at 1000, roughly as $1/N$. The mass past 1 drops from 6.1% to 0.02%. With little data it is worse: after 2 heads in 2 tosses the Laplace sd is 0.217 against an exact 0.178, and 12% of the Gaussian lies past 1.

Bounds, extra peaks and logistic weights

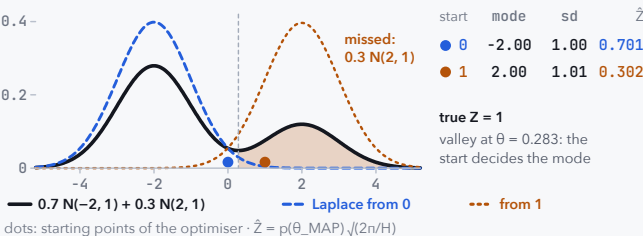
The same recipe on a bounded parameter, a two-peaked target and logistic regression weights.

5 For a bounded parameter, fit in log-odds



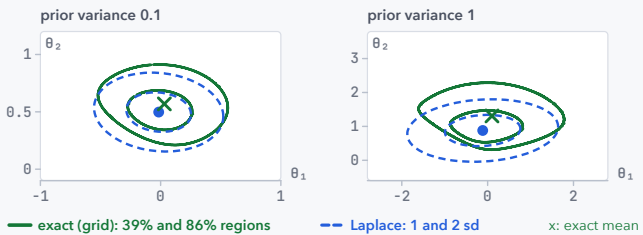
With $\varphi = \log(\theta/(1 - \theta))$ the density of φ is $\propto \theta^{11}(1 - \theta)^3$ (the Jacobian adds $\theta(1 - \theta)$), with mode $\theta = 11/14$ and curvature $14\theta(1 - \theta) = 2.36$. Mapped back, this Gaussian stays in $(0, 1)$ and has KL 0.039: the result depends on the parameterisation. The notebook's second coin works on this scale ($\mathcal{N}(-2, 1)$ prior); its printed MAP 0.5143 is an unconverged Adam iterate; Newton gives $\varphi = 0.583$.

6 Two peaks: Laplace describes only one



The target $0.7\mathcal{N}(-2, 1) + 0.3\mathcal{N}(2, 1)$ has $Z = 1$. From $\theta = 0$ the optimiser reaches -2.00 with $H = 1.00$, so the fit $\mathcal{N}(-2.00, 1.00^2)$ misses the 30% at $+2$; a start right of the valley at 0.283 finds the other peak. $\hat{Z}$ is 0.701 or 0.302. Sampling (next sheet) can visit both.

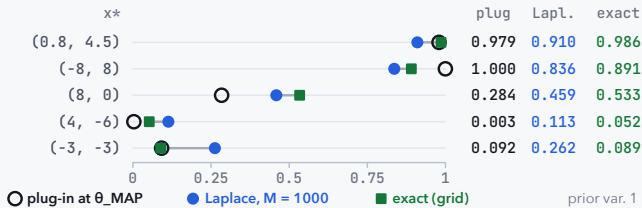
7 Logistic weights: $H = X^T S X + I/\sigma^2$



Prior $\mathcal{N}(0, \sigma^2 I)$, $\mu_i = \sigma(\theta^T x_i)$, $S = \text{diag}(\mu_i(1 - \mu_i))$:
 $\nabla f = X^T(\mu - y) + \theta/\sigma^2$, $H = X^T S X + I/\sigma^2$

Newton's method uses this H at each step. On the 20 points of the Bayesian logistic regression sheet the fit is close at $\sigma^2 = 0.1$. At $\sigma^2 = 1$ it sits at $(-0.12, 0.88)$, while the exact posterior leans up to a mean of $(0.10, 1.31)$.

8 Predict by averaging the sigmoid over draws



$$p(y^* = 1 \mid x^*, D) \approx \frac{1}{M} \sum_{m=1}^M \sigma(\theta_m^T x^*), \quad \theta_m \sim \mathcal{N}(\theta_{\text{MAP}}, H^{-1})$$

with $M = 1000$. Averaging pulls points far from the data toward 0.5, as the exact posterior does: at $(8, 0)$ the plug-in 0.284 becomes 0.459 (exact 0.533). At $(-3, -3)$ the symmetric Gaussian overshoots, 0.262 against 0.089. MacKay's closed form (not in the slides) $\sigma(a/\sqrt{1 + \pi s^2/8})$, $a = \theta_{\text{MAP}}^T x^*$, $s^2 = x^{*T} H^{-1} x^*$, gives 0.449 at $(8, 0)$.

9 The normaliser estimates the log evidence

case	exact	Laplace	gap	gap (Laplace - exact)
coin, 2 of 2 heads	-1.204	-1.069	+0.135	
coin, 9 of 10	-4.963	-4.926	+0.037	
coin, 90 of 100	-35.657	-35.655	+0.002	
log-odds coin	-8.939	-8.951	-0.012	
logistic, var 0.1	-3.153	-3.142	+0.011	
logistic, var 1	-1.581	-1.519	+0.062	
logistic, var 100	-1.098	-1.127	-0.030	
two modes	0.000	-0.355	-0.355	

log $p(D)$ in nats · exact: Beta functions, fine grids, or $Z = 1$ · bars clipped at ± 0.4

The log of the boxed Z in panel 2:

$$\log p(D) \approx \log p(D, \theta_{\text{MAP}}) + \frac{d}{2} \log 2\pi - \frac{1}{2} \log |H|$$

For the coin the gap shrinks with data, 0.135, 0.037 and 0.002 nats after 2, 10 and 100 tosses. With two peaks it loses a whole peak: $\log 0.701 = -0.355$.

10 Pause and think

Why centre on the MAP and not the mean? The mean needs Z ; the MAP needs only f , the loss you already minimise.

What if H is singular? The variance is infinite along the flat direction; a proper prior adds I/σ^2 .

Is this the Laplace prior? No. That is the L1 penalty of the MAP sheet; the two share a name only.

Any intractable function? The slides claim it; it needs one dominant, smooth peak.

$$\theta_{\text{MAP}} = \text{argmin } f \cdot H = \nabla^2 f(\theta_{\text{MAP}}) \cdot p(\theta \mid D) \approx \mathcal{N}(\theta_{\text{MAP}}, H^{-1}) \cdot \text{sharp peak, small variance} \cdot \text{good for one peak}$$

Source: slides "Laplace approximation" and "Bayesian logistic regression" and their notebooks, nipunbatra.github.io/pml-teaching. Numbers recomputed in NumPy (closed forms, finite differences, Newton, grids, seed 0); printed notebook results are quoted and replayed. The notebook and the slides' coin figures use 9 heads, 1 tail; the slide's list D has 8 and 1 (Beta(10, 3), MAP 0.818, sd 0.116). Fixed: the 2D "cross section at $x_1 = 0$ " is at $x_1 = -2.44$ (grid row 10); the log-odds coin MAP is 0.583; the logistic notebook's Laplace covariance at $\sigma^2 = 0.01$ counts the likelihood Hessian 20 times (broadcasting) and should be $\begin{pmatrix} 0.0079 & -0.0016 \\ -0.0016 & 0.0048 \end{pmatrix}$; the slides' Z uses D for data and dimension.